

16. _____ require intelligence to handle tasks such as object manipulation and navigation, along with sub-problems of localization, motion planning and mapping.

- a. Machine learning
- b. Robotics
- c. Virtual reality
- d. Augmented reality

17. The most common and currently available AI system is _____

- a. General Intelligence
- b. Narrow Intelligence
- c. Super Intelligence
- d. Theory of Mind

18. Which of the following does not belong to components of AI system?

- a. Models
- b. Data
- c. Hardware
- d. Programming languages

19. Which of the following AI system has beaten the world's chess champion?

- a. AlphaGo
- b. Deep Blue
- c. Siri
- d. Watson

20. All are AI applications in the agriculture sector, except _____.

- a. See & Spray Robot
- b. Harvest CROO Robotics
- c. Crop and Soil Monitoring
- d. IBM Watson analysis

Part III: Short Answer (1 pt each)

1. _____ is a discipline concerned with the design, evaluation and implementation of interactive computing systems for human use and with the study of major phenomena surrounding them.
2. A technology that enable machines to analyze, understand and manipulate images and video is known as _____.
3. _____ is the science and engineering of making machines to think and act humanly.
4. The two method of computing for Big Data Processing are _____ and _____.
5. The "three Vs. of big data" are _____, _____, and _____.

6. The bi-direction information flow between a human brain and a machine is referred to as
- Neuro-modulation
 - Social Computing
 - Human Machine Interface
 - Brain Computer Interface
7. Agricultural mass crop production technologies were invented during _____ period
- Industrial revolution
 - First Agricultural revolution
 - Second Agricultural revolution
 - Third Agricultural revolution
8. Poorly designed human-machine interfaces can lead to many unexpected problems. All can be classic example, except _____
- Three Mile Island accident in USA
 - A nuclear meltdown accident
 - Accidents in aviation industries
 - Explanation of Chabots
9. All are characteristics that determine the quality of information, except _____
- Timely
 - Accuracy
 - Completeness
 - Inconsistency
 - All of the above
10. The goal of most big data systems is _____
- To save cost
 - To reduce complexity
 - To surface insight
 - To secure data
11. If you select one photo in your phone and see details like the size, type, location, and image captured date, this is _____
- an integer data
 - a metadata
 - a semi-structured data
 - a unstructured data
12. Which of the following is wrong about data type?
- A data type is an attribute of data in integers, Booleans, and floating-point number.
 - A data type is about the structure of data such as structured, unstructured, and semi-structured.
 - A data type is the structure of the data for a computer programming.
 - A data type is data about data in metadata
13. What are the different activities of Big-data life cycle?
- Ingesting data, persisting data, computing and analyzing data, and visualizing results.
 - Acquisition, analysis, curation, storage, and usage.
 - Veracity, variability, and value.
 - Input, processing, and output.
14. With respect to "commuting and analyzing Big data", one of the following tool is different
- Apache Storm
 - Apache Hadoop's MapReduce
 - Apache Flink
 - Apache Spark
15. Which of the following are the applications of Chatbots?
- Google Assistance
 - Google Home
 - Siri in iphone
 - None of the above

Addis Ababa University
College of Natural and Computational Sciences
Introduction to Emerging Technologies (EmTe1011) for Social Science
Mid-term exam (total point 30%)

Date: December 23, 2019
Time allotted: 1hr.

Name: _____ Id no.: _____ Section: _____

Select campus/school: CSS | CHLSJC | CEBS | CBE-FBE | CBE-Commerce | CLGS | Art | Yared

General instruction: The exam contains total of 30 questions provided in three parts: true/false, multiple choice, and short answer. All answers should be written on the answer sheet included at the bottom of page 3. Switch-off mobile phone.

Part I: True or False (1 pt each)

1. When apps move from the cloud back to on-premises it is known as cloud repatriation.
2. Data curation involves key activities concerned with annotating data so that it becomes easily accessible, discoverable and fit for purpose.
3. Real-time processing is most useful when dealing with very large datasets that require quite a bit of computation.
4. Sophia has been designed by Ethiopian Engineers and is an example of limited memory AI type.
5. Self-driving cars are example of reactive machines artificial intelligence.

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Part II: Multiple Choice (1 pt each)

1. Which of the following is regarded as smart revolution?
 - a. Information revolution
 - b. Agriculture revolution
 - c. Industrial revolution (Second revolution)
 - d. Knowledge revolution
2. Which of the following factor is not considered as a driver of information revolution?
 - a. The growth of information-based occupation
 - b. The invention artificial intelligence
 - c. The widespread of cheap personal computers
 - d. The advent of the Internet
3. A device that provides interfacing, data communication, signal processing and other key similar functionalities is _____.
 - a. Microprocessor
 - b. Memory device
 - c. Logic device
 - d. Network device
4. Which of the following is not a future trend of networks?
 - a. 5G technology
 - b. Rise of centralization
 - c. Embedded computation
 - d. Network developments in edge computing
5. Stage of industrial revolution characterized by the use of IoT, AI, and Big Data Technologies to the industry that will allow the intelligent production.
 - a. Industry 1.0
 - b. Industry 2.0
 - c. Industry 3.0
 - d. Industry 4.0

Part IV Give the short answer for the following questions.

1. List out and discuss the characteristics of Big Data? (4pts)

A. _____

B. _____

C. _____

D. _____

2. List the characteristics of Hadoop(3pts)

A. _____

B. _____

C. _____

D. _____

3. List the advantage of Artificial Intelligence at least five? (5pts)

A. _____

B. _____

C. _____

D. _____

E. _____

Part V Give brief description for the following questions

1. Give brief Explanation about the Big Data?(4pts)

2. Why we need artificial intelligence? (4pts)

15. One of the following is the type of Artificial intelligence based on the functionality except one?

- A. Reactive Machines
- B. Theory of Mind
- C. Self-Awareness
- D. General AI

Part III matching (1pts each questions)

COLUMN "A"

1. Super Artificial Intelligence
2. Industrial Revolution (IR 3.0)
3. Narrow Artificial Intelligence
4. Semi-structured Data
5. General Artificial Intelligence
6. Structured data.
7. Industrial Revolution (IR 2.0)
8. Metadata.
9. Data value Chain.
10. Machine perception

COLUMN "B"

- A. SQL Database, MS-Access and MS-Excel
- B. To describe the information flow within a big data system.
- C. Perform any task better than a human with cognitive properties.
- D. The last category of data type
- E. It used to the ability to use input from sensors like Camera and Microphone
- F. Digital Revolution
- G. The most common and currently available Artificial Intelligence
- H. XML and JSON
- I. Technological Revolution
- J. Perform any intellectual task with efficiency like a human.

7. One of the following is introduced or generated by in second industrial revolution?
- A. Internet C. Electrical power and Telephones.
B. Telegraph D. B&C E. A&B
8. One of the following is the interconnecting of unlimited number computers, connectivity devices and cables those are used to share hardware, software and data from one device to another's.
- A. Logic devices C. Memory devices
B. Network D. Microprocessor
9. One of the following is not the high profile of Artificial Intelligence?
- A. Prediction of judicial decisions
B. Self-driving cars
C. Spam filtering
D. Proving mathematical theorems
E. None
F. Except "C"
10. One of the following is the re-structuring or re-ordering of data by people or machines to increase their usefulness and add values for a particular purpose.
- A. Data analyzing C. Data Accessing
B. Data processing D. Data Requirement
11. One of the following is the main goal of Artificial Intelligence?
- A. Solving the difficult problem
B. Replicate human intelligence
C. learn new things by itself
D. All E. None
12. One of the following is not common example of data type?
- A. Integer C. Text
B. Float D. variable
13. _____ is a general term that describes the delivery of on-demand services, usually through the internet, on a pay-per-use basis.
- A. Check-Box C. Augmented Reality
B. Cloud Computing D. Mixed Reality
14. One of the following is true about Artificial Intelligence in Data Security?
- A. The security of data is crucial for every company and cyber-attacks are growing very rapidly in the digital world
B. AI is helping shoppers to discover associated products with recommended size, color, or even brand
C. AI can automate grading so that the tutor can have more time to teach
D. All

Part I say true or false if the statement is correct the answer will be true otherwise false (1pts each)

1. Emerging technology is a term generally used to describe a new technology and existing technology.
2. The quaternary industrial revolution provide a service like teaching and nursing.
3. Artificial intelligence is the one main elements that give life to Autonomous Vehicles and Automated Robots.
4. Data is information that has been processed in some way.
5. An intelligent agent is software that an assists you, or acts on your behalf, in performing repetitive computer-related tasks.

Part II choose the best answer from the given alternatives of the following questions (1pts each)

1. Which protocol is used to link all the devices in the IoT (Internet of Thing)?
A. TCP/IP C. UDP
B. Network D. HTTP
2. What will Security provide?
A. Saves time and cost C. Long term security
B. Product life span increases D. Secure remote management
3. All of the following accurately describe Hadoop, EXCEPT
A. Open-source C. Java-based
B. Real-time D. Distributed computing approach
4. Hadoop is a framework that works with a variety of related tools. Common cohorts include.
A. MapReduce, Hive and HBase
B. MapReduce, MySQL and Google Apps
C. MapReduce, Hummer and Iguana
D. MapReduce, Heron and Trumpet
5. One of the following is the current emerging technology except one?
A. Artificial Intelligence C. Block-chain
B. Cloud computing D. None
6. One of the following industrial revolutions fundamentally changed and transfer the world around us into modern society.
A. The age of science and mass production
B. The rise of digital technology
C. Smart and autonomous systems fueled by data and machine learning.
D. All

12. E-12
10.18
12.18

27. Which of the following is true about the proponents of the theories of ethnicity?

- A. For instrumentalists, ethnicity is fixed at birth. ✗
- B. For constructivists, ethnicity is created & manipulated by elites for political gains. ✗
- C. For the primordialists, ethnicity is natural and unchangeable. ✓
- D. For the primordialists, one can have many ethnic identities. ✗
- E. For instrumentalists, one's ethnicity is determined by ancestry. ✗

A 28. Which of the following is related to factors that drive child marriage?

- A. The value attached to chastity.
- B. Government's need to educate all its citizens. ✗
- C. Political calculation for accessing and gaining resources. ✗
- D. Awareness of females that they should marry at the age of 18 or above. ✗
- E. Parents' encouragement of their female children to continue their schooling. ✗

D 29. Which of the following practices contradicts with human rights?

- A. Providing equal job opportunities for both men and women.
- B. Marriage arrangement with some occupational groups.
- C. Using tools made by craft workers like tanners, potters and weavers.
- D. Practicing early or child marriage in all societies.
- E. Discouraging female genital mutilation or cutting.

E 30. Which of the following is not included in the definition of an ethnic group?

- A. Sense of solidarity
- B. Shared historical memory
- C. Common culture
- D. Common ancestry
- E. Sense of differentiation

C 31. Which of the following statement matches the idea that craft workers lead a life of paradoxes?

- A. Craftwork is a skill which is naturally impure.
- B. Craft workers prefer to live at the outskirts of a village.
- C. Craft workers are marginalized; yet their products are used by others.
- D. Craft workers are so much appreciated in the society for what they produce.
- E. Craft workers do not want to intermarry with other groups.